Supplementary Materials:
mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA

S1. Hyperparameters

S1.1 Pre-training

Table 1: Pre-training hyperparameters for Phase 1 and Phase 2.

Hyperparameter	**Phase 1**	**Phase 2**
Training steps	50,000	25,000
Warmup steps	2,000	500
Peak learning rate	1×10^{-4}	3×10^{-5}
LR schedule	Cosine decay	Cosine decay
Batch size (effective)	128	128
Gradient accumulation steps	8	8
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)	AdamW
Weight decay	0.01	0.01
Gradient clipping	1.0	1.0
MLM masking probability	0.15	0.15
Heteroplasmy loss weight (λ_{het})	0.0	0.3
Checkpoint rotation	Keep last 3	Keep last 3
Gradient checkpointing	Yes	Yes

S1.2 Model Architecture

S1.3 Fine-tuning Hyperparameters

S2. Mathematical Derivation: Circular Positional Encoding

S2.1 Standard sinusoidal PE

In the original Transformer vaswani2017attention, position p is encoded as:

$$\mathrm{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d}}\right) \tag{1}$$

$$\mathrm{PE}(p, 2i+1) = \cos\left(\frac{p}{10000^{2i/d}}\right) \tag{2}$$

For a sequence of length L, $\mathrm{PE}(0, \cdot) \neq \mathrm{PE}(L-1, \cdot)$ in general. As $d \to \infty$, the angle between $\mathrm{PE}(0)$ and $\mathrm{PE}(L-1)$ grows monotonically with L.

Table 2: mtDNA-FM architecture hyperparameters.

Parameter	**Value**
Hidden size (d)	256
Number of transformer layers	6
Number of attention heads	8
Feed-forward intermediate size	1,024
Attention head dimension	32
Dropout	0.1
Attention dropout	0.1
Max position embeddings	514 (512 + [CLS] + [SEP])
Vocabulary size	4,102
Activation function	GELU
Layer normalization ϵ	10^{-12}
Total parameters	≈5.8M
Positional encoding	Circular sinusoidal (fixed buffer)
Genome length for circular PE	16,569 bp (rCRS)

Table 3: LoRA fine-tuning hyperparameters for each downstream task.

Hyperparameter	**Haplogroup**	**Pathogenicity**	**Heteroplasmy**
LoRA rank (r)	8	4	4
LoRA α	16	8	8
LoRA dropout	0.05	0.05	0.05
LoRA target modules	Q, V	Q, V	Q, V
Learning rate	1×10^{-3}	5×10^{-4}	5×10^{-4}
Epochs	20	30	30
Batch size	32	16	16
Loss function	Cross-entropy	BCE ($w_+ = 2.5$)	Huber ($\delta = 1.0$)
Weight decay	0.0	0.1	0.1
Input representation	CLS token	Variant position token	CLS token

S2.2 Circular modification

We replace p with the circular angle $\phi_p = 2\pi p / L_{\text{genome}}$:

$$\text{PE}_{\text{circ}}(p, 2i) = \sin\left(\frac{2\pi p / L_{\text{genome}}}{10000^{2i/d}}\right) \tag{3}$$

$$\text{PE}_{\text{circ}}(p, 2i+1) = \cos\left(\frac{2\pi p / L_{\text{genome}}}{10000^{2i/d}}\right) \tag{4}$$

At $p = L_{\text{genome}}$:

$$\phi_{L_{\text{genome}}} = \frac{2\pi L_{\text{genome}}}{L_{\text{genome}}} = 2\pi$$

and $\sin(2\pi) = \sin(0) = 0$, $\cos(2\pi) = \cos(0) = 1$.

Therefore $\text{PE}_{\text{circ}}(L_{\text{genome}}, \cdot) = \text{PE}_{\text{circ}}(0, \cdot)$, which is the desired circular continuity property.

S2.3 Properties

- *Uniqueness:* Positions p and p' with $|p - p'| < L_{\text{genome}}/2$ receive distinct encodings (the mapping $p \mapsto \phi_p$ is injective on $[0, L_{\text{genome}})$).

- *Smoothness at junction:* $\|\mathrm{PE}_{\mathrm{circ}}(p) - \mathrm{PE}_{\mathrm{circ}}(p+1)\|$ is constant across all positions including $p = L_{\mathrm{genome}} - 1$ (the circular junction), whereas for standard sinusoidal PE this distance jumps at position 0.
- *Non-learnable:* Implemented as `nn.Buffer`, not `nn.Parameter`; gradients do not flow through the positional encoding.

S3. Per-class Haplogroup Metrics

S3.1 Zero-shot 5-NN (26-class evaluation)

Per-class precision, recall, and F1 from zero-shot cosine 5-nearest-neighbour classification on 757 test sequences across 26 major PhyloTree haplogroup classes. No haplogroup labels were used in pre-training. Sequences from NCBI haplogroup-labeled human mitochondrial genomes (15,364 sequences; stratified 80/10/10 split, `random_state=42`).

Table 4: Per-class zero-shot 5-NN haplogroup metrics (757 test queries, 1,509 train library sequences, cosine distance). No haplogroup labels used in pre-training. Classes with $N < 10$ test sequences (E=4, L5=4, G=10, L4=11, V=11) have high per-class metric variance; interpret their F1 scores with caution.

Haplogroup	**Precision**	**Recall**	**F1**	**N test**
A	0.218	0.475	0.299	40
B	0.667	0.800	0.727	40
C	0.667	0.600	0.632	40
D	0.068	0.100	0.081	40
E	0.273	0.750	0.400	4
F	0.774	0.667	0.716	36
G	0.098	0.400	0.157	10
H	0.061	0.050	0.055	40
HV	0.036	0.040	0.038	25
I	0.762	0.800	0.780	40
J	0.185	0.125	0.149	40
K	0.182	0.150	0.164	40
L0	0.857	0.750	0.800	40
L1	0.727	0.571	0.640	28
L2	0.861	0.775	0.816	40
L3	0.485	0.400	0.438	40
L4	0.474	0.818	0.600	11
L5	1.000	0.750	0.857	4
M	0.412	0.175	0.246	40
N	0.133	0.095	0.111	21
R	0.857	0.316	0.462	19
T	0.133	0.050	0.073	40
U	0.389	0.175	0.241	40
V	0.056	0.091	0.069	11
W	0.000	0.000	0.000	16
X	0.071	0.083	0.077	12
Macro avg	**0.3703**			757

S3.2 LoRA Fine-tuning (CPU, 2 epochs)

Results from LoRA fine-tuning under CPU compute constraints (2 epochs; class collapse to haplogroups D, G, R). Non-zero recall for D (24.4%), G (62.3%), and R (13.6%) reflects the partial model: these three haplogroups have the largest training window counts and were the only predicted classes. All other classes have zero recall due to class collapse. See Section 4.2 of the main text for full discussion.

Table 5: Per-class haplogroup classification metrics from LoRA fine-tuning (CPU, 2 epochs). N test refers to the number of test windows (not sequences) per class; each genome contributes ≈65 windows.

Haplogroup	**Precision**	**Recall**	**F1**	**N windows**
A	0.000	0.000	0.000	3,250
B	0.000	0.000	0.000	7,215
C	0.000	0.000	0.000	4,160
D	0.040	0.244	0.069	2,925
E	0.000	0.000	0.000	390
F	0.000	0.000	0.000	2,145
G	0.011	0.623	0.021	780
H	0.000	0.000	0.000	9,945
HV	0.000	0.000	0.000	845
I	0.000	0.000	0.000	1,170
J	0.000	0.000	0.000	2,470
K	0.000	0.000	0.000	1,300
L0	0.000	0.000	0.000	3,575
L1	0.000	0.000	0.000	1,820
L2	0.000	0.000	0.000	4,485
L3	0.000	0.000	0.000	5,265
L4	0.000	0.000	0.000	780
L5	0.000	0.000	0.000	260
M	0.000	0.000	0.000	5,265
N	0.000	0.000	0.000	1,170
R	0.015	0.136	0.026	1,040
T	0.000	0.000	0.000	4,160
U	0.000	0.000	0.000	6,760
V	0.000	0.000	0.000	975
W	0.000	0.000	0.000	520
X	0.000	0.000	0.000	585
Macro avg	0.003	0.039	0.004	73,255

S4. Dataset Summary

S5. DVC Pipeline

The complete analysis pipeline is managed with DVC. The following stages are defined:

1. `download_hmtdb`: Download and cache HmtDB sequences
2. `download_ncbi`: Download vertebrate mtDNA from NCBI Entrez
3. `download_variants`: Download gnomAD, ClinVar, PhyloTree variant data
4. `preprocess`: Quality filter, normalize to rCRS length, train/val/test split
5. `build_vocabulary`: Construct 4,102-token k-mer vocabulary
6. `pretrain_phase1`: Phase 1 cross-species pre-training (50k steps)
7. `pretrain_phase2`: Phase 2 human specialization (25k steps)

Table 6: Datasets used in this study.

Dataset	Source	Size	Purpose
HmtDB	clima2017hmtdb	34,975 seqs used (47,000 total in database)	Phase 2 pre-train, fine-t
NCBI vertebrate mtDNA	NCBI Entrez	117,615 seqs	Phase 1 pre-train
PhyloTree	vanoven2009phylotree	26 major groups	Haplogroup labels
gnomAD v3.1 chrM	karczewski2020gnomad	419 benign SNPs (AF >1%)	Benign variants
ClinVar	landrum2018clinvar	118 pathogenic SNPs	Pathogenic variants

8. `finetune_haplogroup`: LoRA fine-tuning for haplogroup classification
9. `finetune_pathogenicity`: LoRA fine-tuning for pathogenicity prediction
10. `finetune_heteroplasmy`: LoRA fine-tuning for heteroplasmy regression
11. `evaluate`: Compute all metrics, generate evaluation reports

Reproduce the full pipeline: `dvc repro`

S6. Ablation Learning Curves

Ablation training curves (circular PE vs. sinusoidal PE vs. learnable PE; two-phase vs. single-phase training; with vs. without heteroplasmy channel) will be included in the next version of this preprint following GPU-accelerated ablation runs.

References